

Institutional Behaviour Research Platform — Project Report

Author: Markandeya Varma **Date:** 23 August 2026 **Repositories:** `MICC` (V1) · `MICCV2` (V2) · `institutional-research` (V3, current)

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1. Introduction

This report describes a research system I have been building since June 2026. The system studies the Indian stock market. Its purpose is to find out whether the buying and selling done by large institutions — mutual funds, foreign investors, insurance companies — can tell us anything useful about where share prices go next.

The project has gone through three versions in about eight weeks. This report covers all three. It explains what each version did, what it produced, why the first two were stopped, and what the current version is doing.

I want to say clearly at the start what the honest position is, because the rest of the report only makes sense with this in view:

The system has not found a way to make money. Two versions have been built and stopped. The third is about a quarter built. On everything measured so far the likely answer is *"there is no usable edge here"* — and this version is built so that it can say so plainly instead of avoiding it.

That may read like failure. I would argue it is the opposite. The first two versions could not have told me they were failing, and that is why they ran for eight weeks. The third version is built so that it can.

1.1 What the system actually does

Every day, the two Indian stock exchanges publish a list of large trades. When somebody buys or sells more than 0.5% of a company's shares in one day, the exchange must publish it: the date, the company, the name of the buyer or seller, the quantity, and the price. This is public information. Anybody can download it.

The research question is simple to state:

When a large institution buys a share, does that share do better than it otherwise would have? And if so, is the effect big enough and reliable enough to be worth acting on, after costs?

The question is simple. Getting a trustworthy answer is not, and most of this report is about why.

1.2 Reading this report

All numbers in this report were measured from the actual repositories and data on 17–23 August 2026, and re-measured against the rebuilt warehouse on 22–23 August where that changed the answer. Nothing is estimated. Where a number is uncertain I say so. Git commit hashes are given so any claim can be checked.

2. Why I am building this

2.1 The personal reason

I want to learn how to do quantitative financial research properly. Not how to build a trading bot — that is the easy and useless part — but how to tell the difference between a real pattern and a coincidence. That distinction turns out to be the entire subject.

2.2 The intellectual reason

Financial markets produce enormous amounts of data, and almost every pattern found in that data is false. Test ten thousand ideas against past prices and some will look brilliant purely by chance. This is not a technicality. It is the central problem of the field.

The professional literature has developed real tools for this — pre-registration, multiple-testing corrections, out-of-sample testing, power analysis. Most amateur work ignores them completely, finds something that looks wonderful in a backtest, and loses money. I wanted to build something that uses the real tools.

2.3 The specific reason this dataset

Bulk and block deal disclosures are unusually good for this kind of study for three reasons:

1. **They are events with a date.** You know exactly when the information became public, so you can measure what happened afterwards without guessing.
2. **They are free and long.** India has published them since 2006. I have 223,450 bulk deals and 12,430 block deals going back twenty years.
3. **There is a plausible story.** Large institutions have research departments. If anybody has an information advantage, they should. So there is a reason to expect an effect, rather than just data-mining and hoping.

Point 3 matters more than it looks. Searching data with no idea of what you are looking for is how false discoveries are made. Having a reason in advance is what separates research from fishing.

2.4 What I am not building

I want to be exact about this, because it affects the design:

- **No live trading.** No code that places orders exists anywhere in the current repository. There is an automated test that fails the build if anybody adds functions named `place_order` , `modify_order`

or `cancel_order`.

- **No money is at risk.** Nothing is connected to a real trading account.
- **No product.** This is not a startup or a service. It is a research project.

3. The previous versions, and proof of work

Three versions exist. Two are stopped. All three are in Git, so the history below can be independently verified.

3.1 Version 1 — MICC (28 June – 8 July 2026)

Repository: `~/Workspace/MICC`

Measure	Value
Commits	67
Period	28 June 2026 → 8 July 2026 (11 days)
Files	191
Python code	118 files, 23,172 lines
Final commit	<code>e27991f</code> — <i>"Portable path layer + SHP backfill completion + verification hardening"</i>

First and last commits:

```
2026-06-28 2af2cb1 Initial commit
2026-06-28 ce386d8 Clean data-extraction project: data_extraction/ + data_storage/
2026-06-28 5adf4bd Fix all extractors: bootstrap, table creation, dead URLs, parquet
export
...
2026-07-07 9cf8af0 Frontend: Portfolio page, idea-card lifecycle UI, resilience/a11y/perf
+ tests
2026-07-08 e27991f Portable path layer + SHP backfill completion + verification hardening
```

What V1 was for. V1 was a data collection project. Its job was to gather Indian market data from public sources and store it in a usable form. It did that job, and it did it well.

What V1 produced. The lasting output of V1 is a dataset of **116 tables containing 11,276,328 rows, totalling 1.2 GB**. This still exists, and the current version is built on it. The largest tables:

Table	Rows
shp_institutional_summary	3,565,899
shp_promoter_group	2,526,543
window_extremes	1,087,140
shp_category_summary	892,410
pit_universe	359,047
features_monthly	343,963
insider_trading	283,281
bulk_deals	223,450
global_indices_daily	212,955

Why V1 stopped. It did not fail. It was replaced because it had grown into a mixture of data collection, a web frontend and analysis, all in one codebase, with no clear separation. Rather than untangle it, I started V2 and carried the data across. **This was a reasonable decision and the data has proved its value — it is irreplaceable, because the exchanges do not serve twenty years of history on request.**

3.2 Version 2 — MICCV2 (11 July – 13 August 2026)

Repository: ~/Workspace/MICCV2 · **Frozen at tag** frozen-2026-08-16

Measure	Value
Commits	107
Period	11 July 2026 → 13 August 2026 (34 days)
Files	411
Python code	204 files, 35,762 lines
Tests	45 files, 375 test functions
Reports generated	136

What V2 was for. V2 was an attempt to build a complete research and paper trading system: data warehouse, signal engines, a strategy factory that would generate and test candidate strategies automatically, a portfolio simulator, and a dashboard.

The engineering was good. This is worth stating plainly. V2 has a real data warehouse, working data collection, 375 automated tests, and a discipline of generating verification reports. As a piece of software engineering it is competent work.

The research produced nothing. This is also worth stating plainly.

The strategy factory generated and tested candidate strategies for about five weeks. The number promoted to live use was **zero**. This is not my interpretation — the phrase "0 promotions" appears in **18 separate automatically-generated verdict reports** in the repository.

The final scorecard, from `reports/p7_verdict_2026-08-15.md` :

Metric	Portfolio	Nifty 500 TR	NIFTY 50
Sharpe ratio	1.10	0.67	0.39
CAGR	8.1%	9.2%	4.6%
Volatility	7.4%	14.9%	13.7%
Max drawdown	-7.9%	-18.5%	-15.8%

The system's own pass mark was a Sharpe ratio of 1.3. It scored 1.10. Its own report calls this **"BELOW FLOOR"**.

Worse, and to the credit of whoever wrote that report, it says so itself:

"Of 648 book days, 624 are REPLAY (backtest of the champion) and 24 are inside the live-forward window. Of those 24 live-window days, only 9 were CAPTURED LIVE."

So the 1.10 Sharpe is mostly a backtest, and the live evidence is 9 days.

Why V2 was stopped. Three reasons, all found during an audit in August 2026.

Reason 1 — the cost model was wrong. V2 calculated trading costs incorrectly in four separate ways: it charged Securities Transaction Tax on one side of a trade instead of both, used the wrong exchange transaction rate, and applied GST to the wrong base. Total understatement: **10.04 basis points per round trip**.

This is not a rounding error. V2's best seasonal finding, a "turn of month" effect, showed **+3.70 basis points** of profit. With the costs corrected it becomes **-6.36 basis points** — a loss. The system's headline finding was an artefact of its own accounting mistake.

Reason 2 — it could not fail. V2 recorded 22 killed ideas and zero promotions, over about two years of accumulated development effort, and never once concluded that the whole approach might not work. There was no rule anywhere that could stop the project. A system that can only ever say "keep going" is not performing a test.

Reason 3 — documentation had drifted from reality. V2's README described an automated schedule that no longer matched the actual schedule on the machine. Nothing checked, so nobody knew.

The V2 audit is itself part of the work. Finding these three things took several days and required reading the whole system. That audit is why V3 exists and why V3 is designed the way it is.

3.3 Version 3 — institutional-research (16 August 2026 – present)

Repository: `~/Workspace/institutional-research`

Measure	Value
Commits	31
Period	16 August 2026 → present
Python code	6,243 lines
Tests	325, all passing
Decision records	37

V3 is deliberately much smaller than V2 and does much less. It is described in the rest of this report.

3.4 Summary of the three versions

	V1	V2	V3
Duration	11 days	34 days	ongoing
Commits	67	107	30
Lines of Python	23,172	35,762	6,243
Tests	—	375	325
Main output	11.3M rows of data	136 reports, 0 promotions	discipline framework
Status	superseded	frozen	active
Honest verdict	succeeded at its job	engineering good, research empty	too early to say

4. The complete action plan

The plan for V3 is written in three documents totalling about 100,000 characters (docs/plan/PLAN_1_FOUNDATIONS.md , PLAN_2_METHODOLOGY.md , PLAN_3_EXECUTION.md). This section summarises it.

4.1 The core idea

The plan is built around one insight from the V2 audit: **the hard part is not finding patterns, it is not being fooled by them.** So the system is built discipline-first. The rules that stop self-deception were built before the research code, not after.

4.2 The eight phases

Phase	What it does	Status
0	Audit V2, freeze it, decide what to carry forward	done
0.5	Test whether data sources actually work	done
1	Foundations: paths, hashing, migrations, discipline framework	done
2	Data collection: daily automated capture of new deals	partly done
3	Identity layer: match company names to companies	not started
4	Outcomes: calculate what happened after each deal	not started
5	Benchmarks: compare against similar companies	not started
6	The four studies	not started
7	Seasonality rescan	not started
8	Reporting	not started

4.3 Three tracks

The project runs three research tracks in parallel. They share the discipline machinery — registration, decision records, correction for how many things were tried — and almost nothing else, because the statistics of an event, a calendar pattern and a flow series are genuinely different.

	Track D — deals	Track S — combinations	Track F — FII/DII flows
Question	do disclosed institutional trades predict returns?	do <i>any</i> patterns, found by searching, survive out of sample?	does aggregate foreign and domestic institutional flow predict returns?
Data	223,450 bulk + 12,430 block deals, 2006–2026	31.9 million calendar patterns + signal combinations	15,359 rows of derivatives positioning, 2014–2026 · 68 rows of cash flow
Unit	one deal event	one pattern	one trading day
Machinery	built	not built	not built
Honest state	usable data, one study rejected	already scanned once, found nothing	almost no cash data — see below

4.3.1 Track D — the four deal studies

Four studies are registered. They run in the order below, which is descending order of how much each is expected to be worth.

#	Study	The question	Events
1	Consensus	Do three or more unrelated institutions buying the same share within 21 days predict outperformance?	10,098
2	Selling	Do disclosed institutional sales predict underperformance?	34,270
3	Block deals	Do negotiated blocks behave differently from ordinary bulk deals?	12,430
4	Bulk buys	The original premise, run properly	30,771

Why this order. Consensus is the statistically strongest. Judging a single institution is hopeless — one large fund has 80 buys in twenty years, and at that count skill and luck cannot be told apart. A convergence of several institutions needs no individual one of them to be clever.

Selling comes next because 34,270 events have never been examined, and there is a reason to expect more from them: institutions buy for many reasons — money coming in, index tracking, rebalancing — but they sell for fewer.

Blocks are the cleanest data in the project, but there are only about 620 a year.

Bulk buys come last because they already look dead, at $t \approx -0.8$. The study exists to close that honestly rather than leave it informally dismissed.

Study 4 carries a disclosure. An exploratory pass over bulk buys was run on 16 August 2026, before registration. The registry records that it happened and what it found. A registration that hides an earlier look is not a registration.

Only one of the four is on the critical path. The schedule in §9.4 guarantees a single study through both gates; the other three are extensions, and §9.4 fixes in advance the order they are dropped in.

Study 1, Consensus, is the one on the critical path (decision 0031, 23 August). Selling, Blocks and Bulk buys are extensions in that order, with Selling leading them.

That had never actually been decided until now — only implied, and implied two different ways. The plan's critical-path row said "one outcome study" without naming it; its numbering implied Consensus; and this report said Selling, which came from reading the plan's note "institutional selling first" as *first*

study rather than *first among the extensions*. An audit of this report on 22 August found the contradiction, and a grep of every decision record confirmed that none of them resolved it.

It was settled on the reasoning already attached to the ranking: Consensus is the only one of the four whose power does not depend on any single institution being good. The cost is accepted rather than argued away — **Selling is the largest unexamined block of evidence in the project, and it does not run on the critical path.**

The ordering is, for the moment, academic. Every short horizon sits below its detection floor, and the design gate refuses to register a study whose horizons are all blind. Only the twelve-month horizon can conclude, and only by 5% (§8.2), so no study runs until that margin is shown to survive. 0031 settles *which* study, not *when*.

4.3.2 Track S — the search track, and the 31.9 million combinations

This is half the project, and it is the half where it is easiest to fool yourself.

What the 31.9 million are. A "calendar pattern" asks one question: *does this share tend to rise during a particular stretch of the year?* Each pattern is one combination of three things — how long the stretch is, where in the calendar it starts, and which company it applies to. Multiplying those out is where the number comes from:

13 window lengths × 242 starting points × 4,200 companies = 13.2 million
plus index-level variants, four calendar alignments, two ways of measuring
= 31,893,556

V2 already scanned all of them, and found nothing. Its own conclusion:

"Essentially none of it survives contact with its own null. The single best pattern found — a 3-day window that rose in 94.7% of years — sits at the 94th percentile of what randomly rotated data produces, which is to say it is an ordinary result of looking 31.9 million times."

And the part that matters most: **the scan of millions found nothing, while eight carefully-reasoned guesses found two.** Both of those two were then killed by the corrected cost model.

So why do it again? Because V2 only ever asked *"is this better than chance in the data I have?"* It never asked *"does it happen again?"* — which is the only question a pattern claim actually makes. Track S asks the second one: find a pattern in 2005–2015, then require it to repeat in 2016 onward, which it has never seen.

What measuring it honestly revealed

Three findings, each of which changed the design before any code was written.

A calendar pattern happens once a year. So a company with 21 years of history gives 21 observations for any one pattern — total, ever. Split into training and testing, that leaves two to five. With two observations you could only detect an effect of about **50% a year**. Per-company calendar analysis is not difficult; it is arithmetically impossible. And the history is not there either: the median company has **5.5 years** of prices, and only 513 of 4,200 have twenty.

Combining companies does not rescue it. The obvious fix is to pool all 4,200 together. But every company experiences a given calendar day *simultaneously*, so they all share whatever the market did that day. Measured on real data, the average pairwise correlation of daily returns is **+0.235**, which reduces 21,000 apparent observations to about **four** genuinely independent ones.

And the obvious repair turned out to be empty. Subtracting the market's own movement removes the shared part — but if "the market" is defined as the average of those same companies, then the average of what is left is **exactly zero, on every single day.** I had designed a measurement incapable of producing a number, and only found it by checking.

What survived

One approach. Instead of asking "*is this stretch of the calendar good?*", ask **"does it rank companies consistently, and does that ranking hold up in years it has never seen?"** A persistent *ordering* is far harder to produce by luck than a persistent average — chance does not put four thousand things in the same order twice.

Measured, this can detect a signal strength of about **0.014**, against typical real-world values of 0.02 to 0.05. It is the only formulation in this project that can see something real, and that single fact is why the search track is worth building at all.

The second half: combinations of trading signals

Calendar patterns are only one of the two things being searched. The other is **combinations of trading signals** — and these have much better arithmetic, because a signal can fire dozens of times a year per company rather than once.

A "signal" is a simple observable fact about a share on a given day. Seven families are in scope, and **each one has to have a reason stated before it is allowed in:**

Family	Roughly	Why it might work
Momentum	40 variants	prices react to news slowly, so recent movement continues
Reversal	30	whoever supplies liquidity in a panic gets paid for it
Volatility	25	calmer shares have historically returned more than their risk suggests
Volume	25	unusual activity marks unusual attention
Liquidity	20	harder-to-trade shares must offer more to compensate
Seasonal	30	money genuinely does arrive on a calendar — salaries, index changes
Institutional	20	Track D's deal signals, reused as inputs here

These get combined up to three deep, with thresholds — "*momentum in the top fifth, AND volume more than twice normal, AND an institution bought last week*" — which produces hundreds of thousands of candidates.

Requiring a reason for each ingredient is not a formality. The search is deliberately wide, but width is not a licence to include things nobody can explain. A combination that works and cannot be explained is far more likely to be a coincidence than a discovery.

How the testing actually works

This is the part that separates it from V2's approach. Each candidate is judged by repeated rounds of **train on the past, test on what comes next:**

round 1:	learn from 2005–2010	then test on 2010–2012
round 2:	learn from 2005–2011	then test on 2011–2013
round 3:	learn from 2005–2012	then test on 2012–2014
	... and so on, sixteen rounds	

The training window always starts in 2005 and stretches further each time, which mimics how the method would actually have been used: at any point you know everything up to today and nothing after it.

But sixteen rounds are not sixteen pieces of evidence. Because each training window contains almost all of the previous one, consecutive rounds are roughly 95% the same calculation. Counted honestly, sixteen rounds carry about **eight** genuinely independent tests. A second scheme that shuffles which stretches of time are used for testing produces about 120 rounds worth roughly 20.

Every result will report both numbers — the rounds run and the independent tests they actually represent. Quoting the first without the second is one of the easiest ways to make thin evidence look thick, and it is exactly the sort of thing this project exists to avoid.

The deliverable is not a list of patterns

Picking the best of 31.9 million is hopeless — the best of that many coin flips looks impressive too. So the headline result is deliberately a different thing:

Across many training-and-testing rounds, how often does a pattern chosen in training actually win in testing?

If the answer is about half the time, then searching does not work on this data — a real and useful finding that goes well beyond this project. If it is meaningfully better than half, that is a durable discovery about *method*, which is worth far more than any single pattern.

This reframing is what makes searching 31.9 million things defensible rather than reckless: **the size of the search stops being a liability and becomes the measuring instrument.** The wider you search, the more precisely you measure how badly searching overfits.

Honest expectation, stated in advance: most likely the answer is "about half the time", and almost nothing survives. That is the outcome I expect and it is still worth having.

4.3.3 Track F — FII/DII flows, and why it is barely a track yet

Every day the exchange publishes how much foreign institutional investors and domestic institutional investors bought and sold in total. It is the most-watched number in Indian financial media, and the obvious question is whether it predicts anything.

The honest state of this track is that I have almost no data.

Source	Rows	Period	What it actually is
Cash-market flow	68	17 Jun – 8 Jul 2026	the real thing — 22 days
Derivatives positioning	15,359	2014 – 2026	a different measure (see below)

Twenty-two days is not a dataset, and this is the one gap that cannot be solved by working harder. The exchange publishes today's figure and does not serve history, so the only way to obtain it is to collect it going forward, one day at a time. Roughly two years of collection are needed before any study is possible — which is past this project's own deadline. **The daily collector runs for exactly this reason: every day not captured is lost permanently.**

The twelve-year series is not the same measure, and saying so matters. It records how institutions were positioned in *futures and options*, not how much stock they bought. Those two things can move in opposite directions — an institution can buy shares while hedging with futures. Treating

one as a proxy for the other is the sort of quiet substitution that produces a confident wrong answer, so every study using it will state plainly that it measures derivatives positioning.

What this track can honestly do now: ask whether *derivatives positioning* predicts returns, using twelve years of real data, clearly labelled as a different question from the cash-flow one. That study is possible today.

What it cannot do: answer the question people actually mean by "FII/DII flows" — not before roughly 2028.

This is the clearest example in the project of a limit set by data rather than by effort or method, and it is stated here rather than discovered later.

4.4 The rules every study must follow

These are the heart of the plan.

Rule 1 — Write down the answer you will accept, before you look. Before a study runs, its hypothesis, method, and pass mark are written to a database and locked. A database trigger physically refuses to let them be changed afterwards. This is verified by trying to change one and being blocked.

Rule 2 — Exploring must be free. The data is split into three parts: 30% for free exploration, 20% for choosing between candidates, 50% for final confirmation. You may look at the exploration part as much as you like at no cost. The confirmation part may be used once per registered study, enforced by code that raises an error.

Rule 3 — Count everything you looked at. If you test 171 ideas, the best one will look good by luck alone. The system calculates how good: with 171 attempts, pure noise produces a best result of about $t = 2.94$, so the bar is set at $t = 3.71$. With 31.9 million attempts the bar rises past $t = 7$, and for patterns measured on few observations it rises further still. These figures were themselves corrected on 18 August — see §7.6.

Rule 4 — Check the same list of confusions every time. Ten standard alternative explanations are checked automatically for every study. Any that do not apply must be dismissed in writing.

Rule 5 — An event study is not a strategy test. A finding must pass two separate gates: does the event move prices, *and* does an actual portfolio built on it beat the same portfolio without it, after costs. Section 7.3 explains why this rule exists.

Rule 6 — The project can be abandoned. If three of the four studies fail, or if nothing has passed by 28 February 2027, the conclusion is written up as a negative result and the project stops.

5. The data warehouse

5.1 Design principles

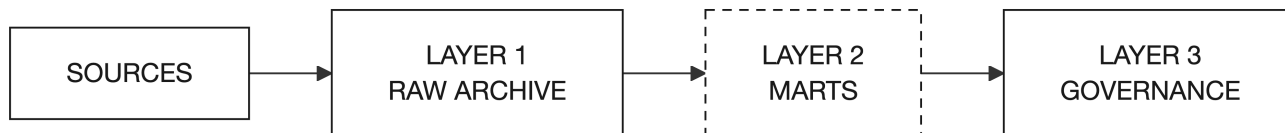
Raw data is never changed or deleted. Files are downloaded, hashed, stored compressed, and never touched again. If one is read wrongly today, it can be read again tomorrow. V2 kept nothing raw, so its 1.2 GB could never be re-derived.

Every day not collected is lost forever. The exchange publishes today's deals at one web address, and replaces it tomorrow. The historical service returns an error. So collection is time-critical: about 27 trading days are already permanently missing.

Derived data can always be rebuilt. Anything computed from raw data can be deleted and regenerated. Only the raw layer is precious.

5.2 The layers

Data moves in one direction through three layers.



A solid box exists and runs today. The dashed box does not exist yet.

Layer	What it holds	Status
Sources	The exchange's daily file, fetched at 20:00 IST	working
1 • Raw archive	Every file as downloaded — compressed, hashed, never changed	built
2 • Marts	The seven clean tables the studies would read	not built
3 • Governance	Registered studies, results, and the provenance trail	built

Layer 2 is the gap. The tables are specified in the plan down to the column, but none has been created.

5.3 Honest status of the warehouse

Layer	Designed	Built	Note
Raw archive	yes	yes	9 files captured so far (3 report types × 3 sessions); 48 KB
V1 seed carried across	yes	yes	2.47 GB carried 22 Aug, every file hash-verified (decision 0027)
Price + F&O spines	yes	yes	7,749,148 and 174,272,768 rows; Phase 1 gate passes 9/9 exact
Provenance DAG	yes	yes	commissioned 22 Aug — 9 artefacts, live lineage (was 0 rows for five days)
Marts	yes	no	0 of 7 tables exist as real schemas
Governance	yes	yes	9 tables, live and enforcing

The marts are the gap. Seven tables are described in the plan in detail, with column lists and constraints, but none of them has been created. They exist as prose in a document, not as a database.

5.4 Two data problems found by measurement

These were found by running real queries. Both are serious and neither was in the plan.

Problem 1 — 8% of deals refer to shares with no price history at all.

Measured against the rebuilt price spine on 22 August:

	Rows	Share
Deal rows, bulk and block	235,880	
Match a symbol in the price spine	215,349	91.30%
Rights entitlements, already out of the universe	1,401	
Unresolved	19,130	8.16%

The first measurement of this, on 17 August, put the failure at 34.2% and the plan explained it as **naming mismatches** — the same company under two symbols — which the identity layer would repair. Re-measured against the real spine, both halves of that turned out to be wrong.

The rate is far lower than 34.2%. But of the 680 symbols behind it:

- **o can be recovered by matching on ISIN.**
- **1 is known to any identity master the project holds.**

There is nothing to map them to. The identity layer cannot convert these events, because conversion needs a target that does not exist.

What they are was not established. 634 of the 680 first appear in 2018 or later, and the names — One Click Logistics, Bright Solar, MOS Utility — look like small-company listings on the SME platforms, which the spine does not cover. That explanation is consistent with everything measured and **could not be confirmed**: the seed holds no NSE SME master to check against. So the decision rests on coverage, which is proven, not on the explanation, which is not.

The decision (0032): they leave the universe. An event on a share with no price series has no forward return to measure, whatever the reason for the gap. They are excluded, flagged, and **reported as an explicit exclusion count beside every study rather than silently dropped.**

Two costs are accepted and must be stated wherever the gate is quoted:

1. **The under-5% gate now passes by exclusion, not by resolution.** A gate that passes because the failing rows were removed is not making the same claim as one that passes because they were fixed.
2. **The excluded events are not random.** They concentrate in small, recently listed companies, so this is now explicitly a mainboard universe and any finding generalises only there — and small-company institutional activity is exactly where an edge is most plausible.

Problem 2 — there is no historical industry classification.

The plan's main method for fair comparison requires knowing what industry a company was in at the time of the deal. The only industry data available covers **1,415 companies out of 4,200 (33.7%)**, and it is a single snapshot taken between 30 June and 2 July 2026. Applying 2026 industry labels to a 2005 event means using information that did not exist yet — the exact error the whole project is designed to avoid.

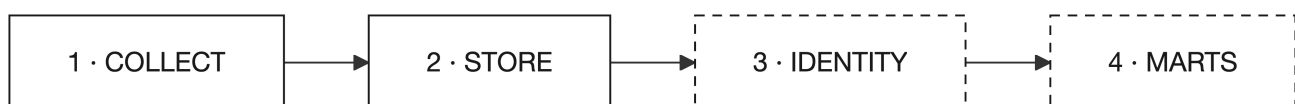
This is currently an unsolved problem, not a task. I do not have a source for it.

6. The system architecture

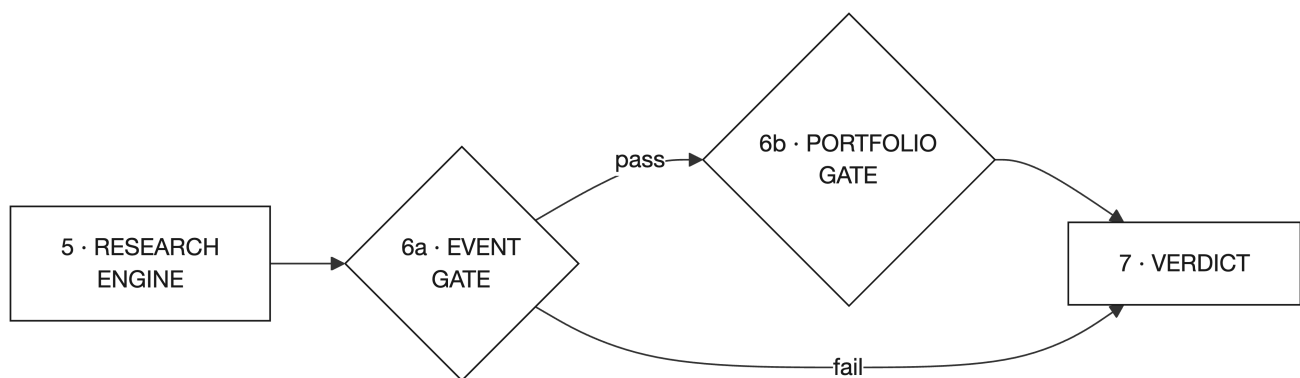
6.1 Overview

The system is a straight line with one side-car. Nothing may skip a stage.

First, the data path. Four stages turn a web page into a trustworthy table.



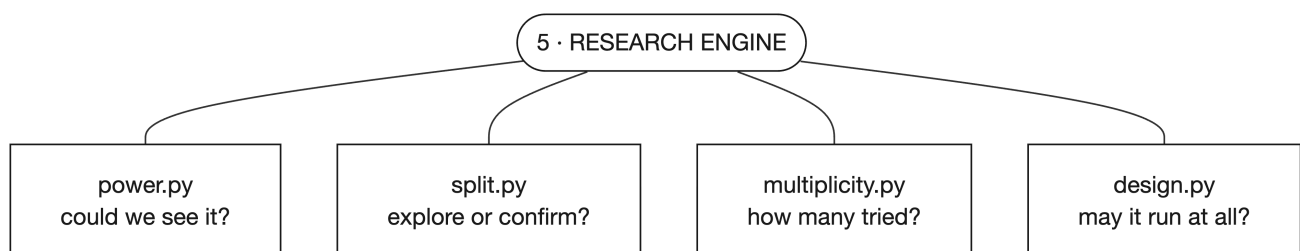
Then, the research path. Three stages decide whether anything in that table is real. A finding must pass both gates; failing the first sends it straight to the verdict.



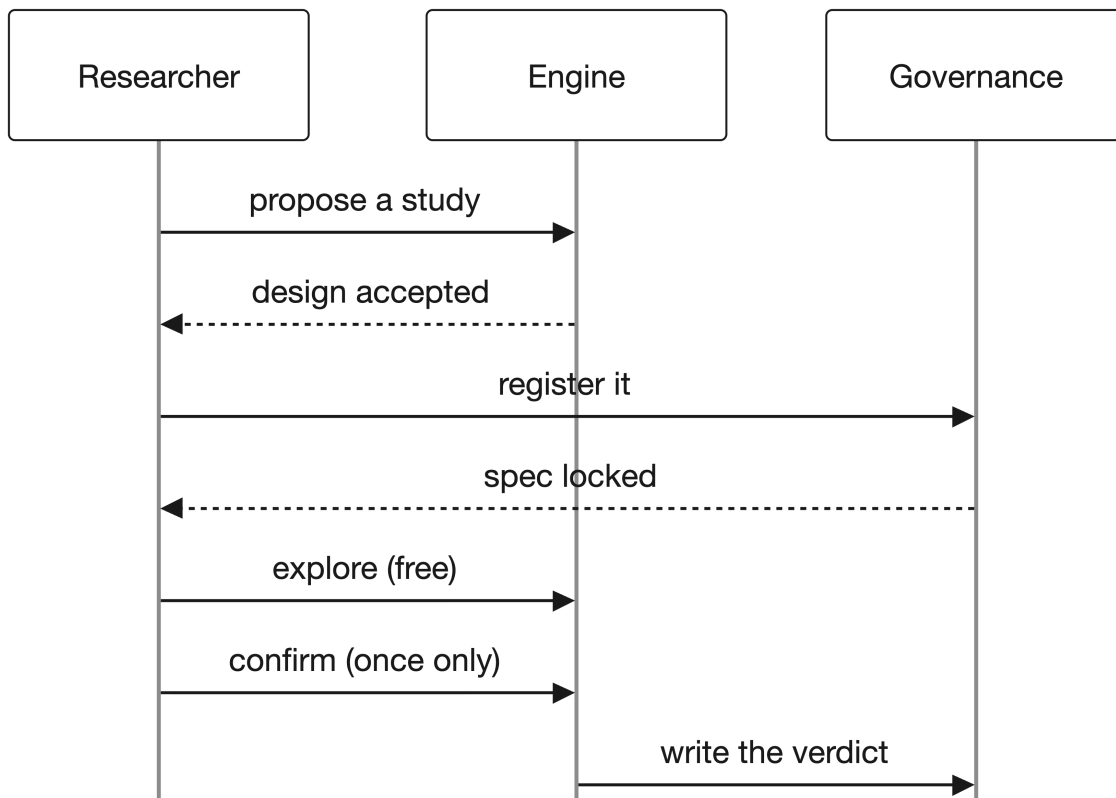
Dashed boxes are not built yet. Governance sits beside all of this: it freezes the question before stage 5 is allowed to read any data.

Stage	What it does	Status
1 · Collect	Fetch today's file from the exchange	part built
2 · Store	Write it once, hash it, never touch it again	built
3 · Identity	Work out which company a name refers to	not built
4 · Marts	Turn raw files into clean tables	not built
5 · Research engine	Decide whether a study may run at all	built
6a · Event gate	Did the deal move the price?	design done
6b · Portfolio gate	Does a real book beat the same book without the rule?	design done
7 · Verdict	PASS, FAIL, or UNDERPOWERED — with its provenance	design done
Governance	Freezes the question before the data is read	built

Stage 5 is the part that is finished, and it is four small modules, each asking one question that V2 never asked:



6.2 How a study flows through the system



The seven steps in words.

1. **Propose.** The researcher describes a study.
2. **Check the design.** The engine asks for a mechanism, a prediction that could fail, and the smallest effect the data could detect. If every horizon is blind, it refuses.
3. **Register.** The hypothesis and the pass mark go into the database.
4. **Lock.** A database trigger refuses any later edit to them.
5. **Explore.** 30% of the data may be looked at freely, at no cost.
6. **Confirm.** The 50% held back may be read once, enforced by code.
7. **Record.** The verdict and the number of things tried are written down, to a table that only ever appends.

6.3 Why the architecture looks like this

Each part exists because of a specific V2 failure.

Component	The V2 failure it prevents
Explicit environment (<code>paths.py</code>)	V2's checker defaulted to "dev", failed there, and rebuilt the <i>production</i> warehouse while reporting on dev
Read-only snapshots	V2 served its dashboard from the live database; 7 of 12 pages returned errors whenever anything was writing
Write-once raw archive	V2's raw folders were all zero bytes — data went straight into the warehouse and could not be re-derived
Locked experiment specs	V2 could not tell a prediction from an explanation made afterwards
Trial counter that sets bars	V2 counted attempts and never used the count for anything
Two gates	V2 had no portfolio test at all, so a finding could pass on event evidence alone
Documentation drift test	V2's README described a schedule that had changed
Provenance graph	V2 could not answer "which numbers came from this data?"

7. What is completed

7.1 The discipline framework — complete and working

This is the substantial achievement so far. 325 automated tests, all passing.

Power analysis (`power.py` , 311 lines). Answers "could this study have seen the effect even if it were there?" before running. If the answer is no, the verdict is UNDERPOWERED — reported as *silence*, not as evidence of absence. This distinction did not exist in V2.

Data split (`split.py` , 341 lines). Divides companies into three groups by a mathematical hash of their ISIN code. Exploration is free; confirmation is enforced by code that raises an error.

A boundary in this rule was undefined until 23 August. An audit of that day's own work found it had never consulted the split at all — roughly a dozen measurements ran across the whole corpus, confirmation names included. It was legitimate: everything computed was *dispersion* — how noisy the data is — and never an effect, and the detectability calculation a study must produce before it may exist has to describe the universe the study will run on, not 30% of it.

But it came out clean by recollection, in a project whose entire argument is that discipline must be mechanical rather than remembered. Decision 0035 draws the line explicitly: **anything that could change a belief about the hypothesis goes through the guard and charges its family; anything that only describes the noise does not.** It is written down now because nothing enforced it before.

The choice of ISIN rather than company symbol was driven by measurement:

Finding	Value
ISINs with more than one symbol over time	276
Those symbols appearing in deal data	459
Deal records affected	26,046 (11.04%)

Real examples: `CADILAH` became `ZYDUSLIFE`; `PRISMCEM` became `PRSMJOHNSN`. If the split used symbols, those companies would sit in the exploration group under one name and the confirmation

group under the other — contaminating the confirmation set for 11% of the data, with nothing looking wrong in any output.

Multiple testing (`multiplicity.py` , 275 lines). Converts "how many things did we try" into a required standard of evidence:

Attempts	Best result from pure noise	Required standard
10	$t = 1.90$	$t = 2.80$
171	$t = 2.94$	$t = 3.71$
1,000	$t = 3.45$	$t = 4.31$
31,893,556	$t = 5.63$	$t = 7.04$

(Corrected 18 August. The figures first published were lower — see §7.6.)

Design gate (`design.py` , 388 lines). Refuses to let a study exist unless it has a mechanism, a prediction that could fail, computed detectability, and a plan for all ten standard confusions.

The test that proves this is worth having is that **it rejects the project's own first experiment**. More on that in 7.3.

7.2 Documentation that cannot silently rot

Thirty-seven decision records, each recording what was decided, by whom, why, **what would reverse it**, and what it costs. An automated test fails the build if any record is missing those fields.

A second test compares the plan documents against the configuration files. On its first run it found three real errors: the plan still advertised abandoned time horizons, never mentioned the portfolio gate, and never mentioned the data split. All three were fixed.

7.3 A completed experiment, and why it matters

One full experiment has been run start to finish: `exp_001` .

The hypothesis. A portfolio of the largest 500 Indian companies which *avoids* any company that had an institutional buy in the last 10 trading days will do better than the same portfolio without that rule.

The sequence, which is the point. The prediction and the pass mark were written to the database and locked in commit `f25608d` . The test was run afterwards. The result was recorded in commit `c31e128` . **The registration is earlier in the Git history than the result**, so it cannot have been written to fit the answer.

What happened at the event level. The effect was real and survived three attempts to kill it:

Check	Result
vs random companies, same dates	-0.860% specific to the event
vs companies of similar volatility	-0.805%, $t = -3.93$
Is it just price reversal?	No — correlation +0.008, pattern not monotonic

What happened at the portfolio level.

Half of the data	Annual difference	t	Verdict
First half	+0.237%	+3.11	FAIL
Held-back half	−0.022%	−0.25	FAIL

Why it died. The rule only excluded **1.2% of companies** at any time. A −0.8% effect applied to 1.2% of a portfolio is about one basis point a month — nothing.

This is the most valuable thing learned so far. Every statistic in the event study was correct. Every one pointed the same way. The study was still useless, and only building the actual portfolio showed it. **An event study is not a strategy test.** The gap between the two is where most amateur work dies.

Under V2's design this would have been recorded as a success.

7.4 A measurement that corrected me

On 17 August I stated that all the project's detectability calculations were too optimistic because they treated companies as independent. **This was wrong**, and I found it wrong within hours by measuring it:

Method	Detectable effect
Treating 16,445 events as independent	0.076% ← never actually done
What the code actually does (monthly grouping)	0.621%
With the correction that <i>was</i> missing	0.660%

The eight-fold penalty was already being applied. The real gap was a different and smaller one — correlation between months, worth 6–27%, not a factor of three.

This is recorded in decision record 0017 rather than quietly edited out. I include it here because a report that only lists successes is not a report.

7.5 Daily data capture — running

A collector runs automatically at 20:00, 22:30 and 08:00 IST. On its first night it captured 100 bulk deals and 4 block deals, verified by hash. Files are compressed, hashed, never overwritten, and duplicates are detected.

Sessions captured so far: 17, 18, 20 August — **19 August is missing**, with no record in the manifest of an attempt. The cause has not yet been diagnosed.

7.6 Five errors of my own, found by verification

On 18 August I was asked simply "*are you sure about the whole thing?*" Checking found four errors in work committed within the previous day. A fifth, of the same kind, was found on 23 August. I record them because a report listing only successes is not a report, and because how they were found matters more than that they existed.

Error 1 — a statistic that measured itself. I reported that removing the market's overall movement from returns made stocks statistically independent, and built a design on it. In fact the calculation I used *forces* that answer no matter what the data contains. Feeding it a strongly-linked dataset and a completely random one produced the same result. It measured the arithmetic of the operation, not the market.

Error 2 — a question with no possible answer. Worse, the way I had defined "market-relative" meant that the quantity I proposed to study is exactly zero for every day by construction. I had designed a measurement incapable of producing a number. The correct approach — comparing *rankings* of stocks rather than their average — was in the plan as one option among several. It is in fact the only one.

Errors 3 and 4 — the significance bar was too low, twice. The tool that sets how strong evidence must be had two independent faults, and both made results *easier* to pass:

Fault	Effect
Measured one direction of movement while being applied to both	every bar understated
Assumed a bell curve where the true distribution has heavier tails	a further 23% understatement for small samples

Corrected, the bar for the deal track rises from $t = 3.62$ to $t = 3.71$. The one completed experiment still clears it, so **no recorded verdict changes**.

Error 5 — the correction for overlapping windows did not reach far enough. Found on 23 August. When the same tool corrects for observations that overlap in time, it chose how far back to look using a standard rule that depends only on *how many* observations there are — and knows nothing about how far each one reaches forward. This project is built almost entirely on overlapping windows: a twelve-month return measured every month shares eleven twelfths of its window with the next one.

On a 252-session label the rule looked back 9 periods against a true overlap of 252, and **understated the inflation fivefold**. Two results reported as POWERED on 22–23 August were artefacts of it, and both fail once the correction reaches the full overlap. The fix makes the look-back at least as long as the label.

It is the same shape as errors 3 and 4 — every one of these five made results *easier* to pass, never harder. That is not chance. An error that makes a finding harder to obtain gets investigated immediately, because it looks like a bug; one that makes a finding easier looks like a result.

Why they survived 146 tests. The tests compared the tool against *its own output*, pinned as expected values. A test asserting that code agrees with itself proves nothing. The replacement tests compare against independent simulation, which is the standard the originals should have met.

What this says about the project. The framework caught none of these — a person asking a direct question did. The safeguards are real and they are not sufficient, and I would rather state that than present a system that appears to police itself.

8. What is going on now

8.1 Measuring whether there is anything to find

The current activity is establishing what size of effect this data can possibly reveal. Results from 17 August, on 16,445 real institutional buy events:

Horizon	Observed effect	Smallest detectable	Conclusion
1 session	+0.691%	0.191%	detectable
5 sessions	−0.010%	0.423%	nothing
10 sessions	−0.603%	0.660%	below the floor
21 sessions	−1.061%	0.968%	detectable but implausibly large

Two things stand out.

The sign flips. Shares rise 0.691% on the first day, then fall, ending 1.061% down after 21 days. This is the classic signature of a temporary price push that then reverses — consistent with the finding that **54.8% of bulk deals are the same party buying and selling on the same day**, which is market-making activity, not investment.

The earlier finding weakens. The 10-session effect of −0.603% is *smaller* than the 0.660% we can reliably detect. The −0.805% found in `exp_001` came from comparing against companies of similar volatility, and it was that tighter comparison that provided the sensitivity — not the effect being robust to any comparison.

The "Conclusion" column above is superseded. It compares each horizon against a single fixed threshold, and §8.2 explains why that comparison was the wrong one. The table is kept because it is what the measurement said at the time and because the sign flip it shows is unaffected.

8.2 The question that affected everything, and what answering it exposed

The system judges whether a study is worth running by comparing what it can detect against what size of effect is realistic. That threshold was a single number — 0.5% per month — compared against every time horizon, though it is only dimensionally correct at about 21 sessions. Comparing a one-day result against a monthly threshold is a units error.

Which fix applies turned on how the effect behaves:

- **If a disclosure causes a single one-off price adjustment**, a fixed threshold is right.
- **If institutions have persistent skill that accrues over time**, the threshold must scale with the horizon.

The owner chose the second on 21 August (decision 0028). An informed institution does not earn its whole edge in the first session and stop. Every short horizon then went blind, and the design gate refuses to register a study where that is true of all of them.

That choice quietly invalidated an earlier one, and it took two days to notice. Decision 0004 had dropped every horizon beyond 21 sessions, reasoning that at 12 months the detectable effect was already 7.38% against a bound of 0.50% — so longer could only be more hopeless. That arithmetic was right *under a fixed bound*. Once the bound scales, it inverts: **the detectable effect grows with the square root of the horizon, while the bound grows in proportion to it**. Longer horizons get easier, not harder. Decision 0028 had made the long horizons viable and dropped them in the same breath.

Re-measured on 23 August — 17,988 eligible buys, round trips and high-frequency participants removed, with the serial-correlation fix of §7.6 applied:

horizon	detectable effect	scaled bound	verdict
21 sessions (1 month)	0.8403%	0.50%	1.68× short
63 sessions (3 months)	1.8506%	1.50%	1.23× short
252 sessions (12 months)	5.7273%	6.00%	within reach

Twelve months is now the primary horizon (decision 0034); the session grid is kept as robustness with its underpowered verdicts intact. It is the only horizon in the entire grid that can reach a conclusion, and a project that cannot conclude anywhere has no path to a result at all.

It must not be read as comfortable. The margin is 5%. It depends on a judgement call about how far the serial-correlation correction should reach — at a more conservative setting the same measurement comes out 1.01× short, which is to say *failing*. It has not yet faced its confound checklist. Twelve-month outcomes also mature slowly: events after August 2025 have no complete window, so the usable events stop roughly a year before the data does and the most recent market regime is invisible to the primary horizon.

And the claim itself is weaker than the one the project set out to test. "An effect that takes a year to appear" is a materially smaller thing than "a disclosure moves the price", even if it clears statistically.

The convenient answer at each step was the other one, and it was rejected each time. Adopting the interpretation that licenses your own conclusions is the failure this project exists to prevent — and 0004 surviving two days past its own invalidation is a reminder that the machinery does not catch this by itself.

9. What is pending

9.1 Build status, measured

Component	Status
Environment, paths, hashing, migrations	built
Discipline framework	built
Daily capture (basic)	built
Trading calendar	not built
V1 data copied into the project	done — 22 Aug, every file hash-verified
Price and F&O spines	built — 7.7M and 174.3M rows
Reconciliation check	passes, 9 of 9 exact
Full collector	not built
Company identity matching	not built
Institution name normalisation	not built
Outcome calculation	not built
Cost model in code	config only
Benchmark construction	config only
The four deal studies (Track D)	not built
Track S — walk-forward folds	not built
Track S — calendar scan	not built
Track S — signal combinations	not built
Track S — the procedure test	not built
Provenance graph populated	9 artefacts, 16 edges

Seven of seven marts still do not exist, and all seven Track S modules are unwritten. What did land on 22 August is the layer beneath them: 4.5 GB of verified data, two spines, and a provenance trail that was empty for its first five days.

Roughly a quarter of the system is built. The completed portion is the framework and the raw layer — not the research machinery, which is the part that would actually answer the question.

9.2 Three things in the plan that could not be built as written

1. No historical industry data (Section 5.4). The plan's main comparison method needs it. It covers 33.7% of companies and is a single 2026 snapshot. There is no known source. **This is a genuine blocker, not a task.**

2. Company matching, now measured and decided. The plan set a 5% limit and gave no method for reaching it. Measured against the rebuilt spine the failure is 8.16%, and none of it is repairable — the shares have no price history anywhere in the project. Decision 0032 excludes them, so the gate passes by exclusion rather than resolution (§5.4). This is no longer an open problem, but the resolution narrows the universe rather than fixing the data.

3. The seasonality rescan has no time estimate. "About three weeks" is an assertion; no single unit of the calculation has ever been timed, and a later decision roughly quadrupled the work.

A fourth has appeared since. **Phase 3, the identity layer, was sized at four weeks on the premise that it would recover the unmatched events.** Decision 0032 established there is nothing to recover, so the phase keeps its other jobs — point-in-time sectors, participant classification, delisting and merger handling — but the four weeks were costed against work that no longer exists and have not been re-cut.

9.3 Risks

Risk	Status
No backup of any kind	open — no remote repository, no external drive. Captured trading days cannot be re-obtained at any price
Data lost each uncollected day	reduced — collector now runs, but ~27 days already lost
BSE data unavailable	open — all routes return errors
Historical backfill impossible	open — service returns error 503
No detectable effect exists	the most likely outcome

9.4 Timeline, and why it had to be rewritten

The plan originally listed nine phases in sequence and stated that no deadline had been set. **That was false** — a deadline of 28 February 2027 had been fixed two days earlier. Measured honestly, at the 2–3 hours a day I actually have:

	Weeks
calendar available	27.9
work planned	26.2
slack	+1.7

Six percent. On a project where **every estimate so far has been wrong**, a 25% overrun — which is optimistic for software — missed the deadline by five weeks.

The fix is not a better estimate. It is deciding **in advance** what gets cut.

The critical path — 14 weeks to one defensible answer

Phase	Weeks	Ends
Warehouse and reconciliation	3	7 Sep 2026
Collection (<i>reduced — the daily capture already runs</i>)	1	14 Sep 2026
Identity layer (<i>4 weeks, now due a re-cut — §9.2</i>)	4	12 Oct 2026
Clean data mart	2	26 Oct 2026
Costs and benchmarks	2	9 Nov 2026
One outcome study — Consensus	2	23 Nov 2026

This lands one week inside the checkpoint, with **13.9 weeks of buffer** — 36% overrun tolerance in place of 6%.

Two reductions make it fit, and neither is guesswork. Collection drops from three weeks to one because the daily capture has been running since 17 August and the raw bytes are already safe; the full parser can wait for a study that needs it. The outcome study drops from four studies to **one**, because one study answered is worth more than four half-answered.

What gets cut, decided now

Extension	Weeks	Cut order
Re-run the first experiment reproducibly	0.2	last
The other three deal studies	3	
The search track	4	
Seasonality	1	
Monitoring and automated reports	2	first

Under deadline pressure, what gets dropped is whatever is least *defended* at that moment — not whatever matters least. Deciding the order now, while nothing is at stake, is the only point at which that decision is honest.

The seasonality decision changed as part of this. I had chosen to rebuild all 31.9 million calendar cells from scratch — three weeks — against advice to validate the existing ones instead. Faced with the schedule arithmetic I reversed that: validation takes one week, and those two weeks are what the buffer is made of. Validation is not "trust the old numbers": a 100,000-cell sample is recomputed from scratch and must match **exactly**, any mismatch escalates automatically back to a full rebuild, and every statistical conclusion is recomputed regardless. What is reused is the raw cell counts, which are cheap to check and expensive to regenerate.

Honest limit. This makes the deadline survivable for the thing that matters — one answer by the checkpoint — but not for everything. Running the full extension list still misses a 25% overrun by 2.4 weeks. That is what the cut order is for.

Where the schedule has actually got to. A plan-versus-actual check on 23 August found three separate drifts, and none of them is visible from the phase table alone:

- **Phase 1 finished about two weeks early.** That is the buffer's only real contribution so far.
- **Phases 2' and 3 are untouched, while parts of 4 and 5 were built ahead of them.** Eligibility filtering and characteristic matching exist because the detectability question demanded them, not because the plan ordered them. Work done out of order is still work done, but it means the phase table overstates how much of the *sequence* has been completed.
- **Phase 3 still carries its four-week estimate**, whose justification decision 0032 removed (§9.2). Nobody has re-cut it.

So the critical path is not behind, but it is not on the track the table describes either.

Date	Milestone
23 November 2026	Critical path complete — first portfolio-gated verdict
30 November 2026	Checkpoint — if no study has reached even the first gate, stop early
28 February 2027	Deadline — if nothing has passed, write the negative result and stop

10. What this is currently used for

I want to be accurate here rather than flattering.

10.1 What it is used for today

Nothing financial. No money is invested. No orders are placed. No recommendations are produced.

The system currently has three real uses:

1. **Daily data capture.** It is preserving Indian institutional deal data that would otherwise be permanently lost. This has value regardless of whether the research succeeds.
2. **A method for testing ideas honestly.** The framework — pre-registration, split data, multiple-testing correction, two gates — is general. It applies to any field where you search data for patterns.
3. **Learning.** I now understand why most quantitative research fails, because I have made the mistakes and measured them.

10.2 What it is not used for

- Not trading, and not connected to any account
- Not advising anyone
- Not producing predictions
- Not producing any validated finding — **zero results have passed both gates**

10.3 What it could be used for if it works

If a study passes both gates and survives the deadline, the output would be a documented, reproducible finding about Indian institutional behaviour, with its provenance recorded. Whether that would be worth trading is a separate question requiring capital, execution and risk management that do not exist here.

On the evidence so far I would put the chance of a tradable finding below 20%. Four things point the wrong way:

- The effect reverses across horizons — up on day one, down by day 21.
- Most bulk-deal activity is market-making, not investment.
- The one completed experiment failed.
- Every short horizon is below its detection floor, leaving one twelve-month horizon that clears its own bound by 5% and has faced no confound checks (§8.2).

10.4 The most likely honest outcome

The most likely result is a well-documented negative: *public institutional deal disclosure in India does not contain a tradable edge for a retail participant after realistic costs.*

That is a legitimate result. It is also unpublishable in most amateur settings, which is precisely why so much amateur work reports positives.

11. Conclusion

11.1 What has been achieved

In roughly eight weeks I have built three versions of a market research system, audited two of them to destruction, and produced:

- A dataset of 11.3 million rows spanning 2005–2026, still in use
- A working data warehouse and collection system (V2), now frozen
- A discipline framework with 325 tests that enforces honest research
- One complete experiment, correctly rejected by its own pre-registered rule
- Thirty-seven decision records with reversal conditions
- Four material measurements that changed the plan: a 10.04 bp cost error, 54.8% market-making contamination, an 8.16% unresolvable share of the deal corpus, and a missing industry history

11.2 What has not been achieved

- **No validated finding.** Zero results have passed both gates.
- **No profitable strategy.** V2 promoted zero strategies in five weeks.
- **The V3 system is roughly a quarter built.** No mart exists, and no study has been run on the data now carried in.
- **Two plan components have no data source.**
- **Only one horizon in the grid can reach a conclusion at all**, and it does so by a 5% margin that has not yet faced its confound checks.

11.3 The honest assessment

If this project is judged on "did you find a way to beat the market", the answer is no, and probably will remain no.

If it is judged on "did you build something that can tell the difference between a real finding and a lucky one", the answer is yes, and that is the harder problem.

The difference between V2 and V3 is not sophistication — V2 has ten times the code. It is that V2 could not fail and V3 can. V2 ran for five weeks producing zero promotions and never concluded anything was wrong. V3 has a written rule that stops the project on 28 February 2027 if nothing works, and a first experiment that was killed by its own pre-registered standard within ninety minutes of looking promising.

11.4 What I have learned

Most patterns in financial data are false. With 171 attempts, noise alone produces a t-statistic of 2.94 — which most people would report as highly significant.

Being able to fail is a feature. V2's real defect was not any single bug. It was that no possible result would have caused it to stop.

Statistics that are individually correct can be collectively useless. exp_001's event study was correct in every particular and the strategy was worthless, because the rule touched 1.2% of the

portfolio.

Measurement beats assumption, including about yourself. Four significant errors were found by measuring rather than reasoning — including one of my own, made and corrected within hours on 17 August.

11.5 Next steps

These are the critical path of §9.4, in order. Everything else is an extension that gets cut before this list does.

1. **Test whether the twelve-month margin survives.** It is the only horizon that can conclude, and it clears its own bound by 5% (§8.2). It has not yet met the confound checklist, era splits, or delisting recovery, and a more conservative serial correction already turns it negative. If it does not hold, no horizon is registrable and the honest conclusion is that Track D cannot conclude. Characteristic matching stays a precondition either way — it cut the detection floor from 1.52% to 1.05% on a single dimension.
 2. **Arrange a backup.** A single disk failure currently loses irreplaceable data, including trading days that cannot be re-obtained at any price. Cheap, and it is the only item here whose cost of neglect is permanent.
 3. **Re-cut the identity layer's four weeks.** That estimate was built on recovering the unmatched events, and decision 0032 established there is nothing to recover. The phase is still needed — point-in-time sectors, participant classification, delistings — but not for the reason it was sized on, and the schedule assumes a number nobody has revisited.
 4. **Establish whether the industry-history problem is solvable.** If it is not, the primary comparison method has to change, and I would rather know that in September than in January.
 5. **Then run one study, properly — Consensus,** fixed by decision 0031: do three or more unrelated institutions buying the same share within 21 sessions predict outperformance? One study through both gates by 23 November is the deliverable. Selling, Blocks and Bulk buys are extensions, in that order. Nothing can be registered until step 1 lands.
-